

ENERGY CONSUMPTION IN BUILDINGS ANALYSIS SYSTEM

Abstract

The Energy Consumption in Buildings Analysis System provides a structured approach to evaluating energy usage across diverse building types. The project focuses on importing and cleaning building energy datasets to ensure accuracy, followed by comparative and temporal analyses. By grouping data according to building type, the system highlights sector-specific consumption patterns, while monthly aggregation reveals seasonal fluctuations in demand. Visualization is central to the methodology, with line charts illustrating energy trends over time, bar charts comparing average consumption across building categories, and pie charts showing the distribution of energy sources. These graphical insights enhance interpretability and support evidence-based decision-making. The workflow is streamlined, beginning with dataset acquisition, progressing through preprocessing and analysis, and culminating in visualization and interpretation. Beyond descriptive analytics, the system emphasizes potential enhancements such as predictive modeling and energy optimization strategies, which can guide sustainable practices and reduce overall consumption. Ultimately, this project delivers a practical framework for monitoring and analyzing building energy usage, enabling stakeholders to identify inefficiencies, promote renewable integration, and contribute to long-term energy sustainability.

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